# 第 4 阶段。重新定位和停用节点

## AFF and FAS Controller Upgrade

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# 第 4 阶段。重新定位和停用节点

## 第 4 阶段。重新定位和停用节点 2

在阶段 4 中，所有非根聚合将从 node2 重新定位到 node3； node2 拥有的非 SAN 数据 LIF 将移至 node3 。您必须记录必要的 node2 信息，然后停用 node2 。

步骤

1. "将非根聚合和 NAS 数据 LIF 从 node2 重新定位到 node3"
2. "停用 node2"

## 将非根聚合和 NAS 数据 LIF 从 node2 重新定位到 node3

在将 node2 替换为 node4 之前，必须将 node2 所拥有的非根聚合重新定位到 node3 。

上一阶段的后处理检查完成后， node2 的资源释放将自动启动。非根聚合和非 SAN 数据 LIF 将从 node2 迁移到 node3 。

在升级操作步骤期间，远程 LIF 处理 SAN LUN 的流量。升级期间，集群或服务运行状况无需移动 SAN LIF 。

迁移聚合和 LIF 后，此操作将暂停以进行验证。在此阶段，您必须验证是否所有非根聚合和非 SAN 数据 LIF 都已迁移到 node3 。

> 不会修改聚合和 LIF 的主所有者；只会修改当前所有者。

步骤

1. 验证所有非根聚合是否均已联机及其在 node3 上的状态：

   ```
   storage aggregate show -node node3 -state online -root false
   ```

   以下示例显示 node2 上的非根聚合处于联机状态：

   ```
   cluster::> storage aggregate show -node node3 state online -root false

   Aggregate      Size         Available   Used%   State   #Vols  Nodes
   RAID     Status
   ----------     ---------    ---------   ------  -----   -----  ------
   -------  ------
   aggr_1         744.9GB      744.8GB      0%     online  5      node2
   raid_dp  normal
   aggr_2         825.0GB      825.0GB      0%     online  1      node2
   raid_dp  normal
   2 entries were displayed.
   ```

   如果 node3 上的聚合脱机或变为外部聚合，请在 node3 上使用以下命令将其联机，每个聚合一次：

```
storage aggregate online -aggregate <aggr_name>
```

2. 在 node3 上使用以下命令并检查输出，以验证 node3 上的所有卷是否联机：

   ```
   volume show -node <node3> -state offline
   ```

   如果 node3 上的任何卷脱机，请在 node3 上使用以下命令将其联机，每个卷一次：

   ```
   volume online -vserver <vserver_name> -volume <volume_name>
   ```

   要与此命令结合使用的 ` <vserver_name>` 可在上一个 `volume show` 命令的输出中找到。

3. 验证 LIF 是否已移至正确的端口且状态为 `up` 。如果任何 LIF 已关闭，请为每个 LIF 输入以下命令，将 LIF 的管理状态设置为 `up` ：

   ```
   network interface modify -vserver <vserver_name> - lif <LIF_name> -home-node
   <node_name> - status-admin up
   ```

4. 如果新硬件上不存在当前托管数据 LIF 的端口，请将其从广播域中删除：

   ```
   network port broadcast-domain remove-ports
   ```

5. 【第 5 步】输入以下命令并检查输出，验证 node2 上是否没有剩余数据 LIF ：

   ```
   network interface show - curr-node <node2>-role data
   ```

6. 如果配置了接口组或 VLAN ，请完成以下子步骤：

   a. 记录 VLAN 和接口组信息，以便在启动 node3 后在 node3 上重新创建 VLAN 和接口组。

   b. 从接口组中删除 VLAN ：

      ```
      network port vlan delete -node <nodename > -port <ifgrp> -vlan-id <vlan_ID>
      ```

   c. 输入以下命令并检查其输出，以检查节点上是否配置了任何接口组：

      ```
      network port ifgrp show -node <node2>-ifgrp <ifgrp_name> -instance
      ```

      系统将显示节点的接口组信息，如以下示例所示：

```
cluster::> network port ifgrp show -node node2 -ifgrp a0a -instance
                 Node: node3
 Interface Group Name: a0a
Distribution Function: ip
        Create Policy: multimode_lacp
          MAC Address: 02:a0:98:17:dc:d4
   Port Participation: partial
        Network Ports: e2c, e2d
             Up Ports: e2c
           Down Ports: e2d
```

a. 如果节点上配置了任何接口组，请记录这些组的名称以及分配给它们的端口，然后输入以下命令删除这些端口，每个端口输入一次：

```
network port ifgrp remove-port -node <nodename > -ifgrp <ifgrp_name> -port
<netport>
```

# 停用 node2

## 要停用 node2 ，您需要正确关闭 node2 并将其从机架或机箱中卸下。

步骤

1. 恢复操作：

   s系统控制器更换恢复

   节点会自动暂停。

升级完成后，您可以停用 node2 。请参见 "停用旧系统"。

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